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AWARD REVIEW

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Chemical studies on bioactive compounds related to higher fungi

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ABSTRACT

Hericium erinaceus (Yamabushitake in Japan) is a well-known edible and medicinal mushroom. We discovered antidementia compounds, hericenones C to H, from the fruiting bodies and erinacine A to I from the cultured mycelia of the fungus. Based on the data of the compounds, several clinical experiments were performed using the fungus. "Fairy rings" is a phenomenon that turfgrass grows more prolific or inhibited than the surrounding area as a ring and then occasionally mushrooms develop on the ring. We found fairy-ring causing principles "fairy chemicals" and the biosynthetic routes of the compounds on the purine metabolic pathway in plants and mushrooms.

Graphical Abstract

Antidementia compounds from *Hericium erinaceus* and fairy chemicals.

Keywords: *Hericium erinaceus*, antidementia compound, fairy rings, 2-zahypoxanthine, fairy chemicals

Since I was appointed to the Department of Agricultural Chemistry, Faculty of Agriculture, Shizuoka University in 1985, as a research assistant, I have consistently conducted chemical and biochemical studies on bioactive compounds related to higher fungi, especially mushrooms. In this review, 2 representative examples among my 35-year-research are introduced.

Antidementia compounds from *Hericium erinaceus* (Yamabushitake)

Hericium erinaceus is a well-known edible and medicinal mushroom in Japan as Yamabushitake, in China as Hou Tou Gu, and in Europe and the United States as Lion's Mane. We discovered

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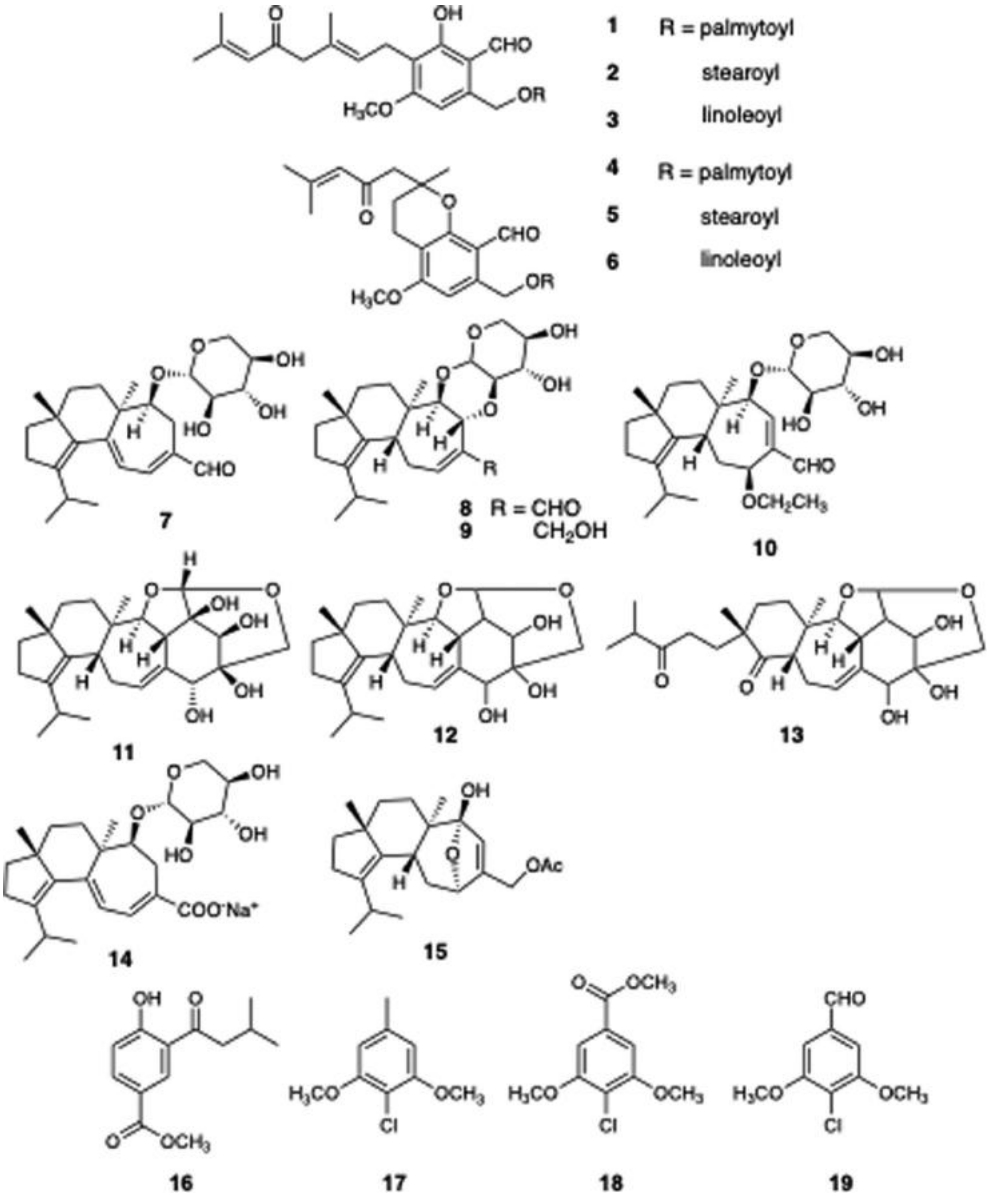

Figure 1. Stimulators of nerve growth factor-biosynthesis and amyloid β-toxicity suppressing compounds from *Hericium erinaceus*.

nerve growth factor (NGF) stimulators from the fruiting bodies of the fungus and named them hericenone C to H (**1-6**) (Kawagishi *et al.* 1991, 1993). Compounds that stimulate the biosynthesis of NGF are considered to be effective against dementia. These compounds were the first NGF stimulators isolated from nature. Later, erinacine A to I (**7-15**) that showed the similar activity were obtained from the cultured mycelia of this fungus (Figure 1) (Kawagishi *et al.* 1994, 1996a,b; Lee *et al.* 2000).

Many types of neurodegenerative diseases including Alzheimer's, Parkinson's, Huntington's, and the prion diseases are due to neuronal cell death and the death is associated with amyloid β-peptide (Aβ). Reduction of endoplasmic reticulum (ER) stress induced by Aβ is one way to cure or prevent such diseases. ER stress–reducing compounds (**16-19**) were obtained from extracts from the scrap cultivation bed of the mushroom (Figure 1) (Ueda *et al.* 2009). Since the cultivation bed is usually discarded by mushroom growers after harvesting the fruiting bodies, another purpose of this research was the efficient use of the waste material.

The effects of hericenone C (**1**), erinacine A (**7**), and the extracts of the fungus on dementia were confirmed in animal experiments (Shimbo, Kawagishi and Yokogoshi 2005; Zhuang *et al.* 2007; Kawagishi and Zhuang 2008; Ratto *et al.* 2019).

Several clinical trials using the fruiting bodies of the fungus have demonstrated the antidementia effect of this fungus (Kasahara, Kaneko and Shimizu 2001; Mori *et al.* 2009; Nagano *et al.* 2010; Ohtomo, Shimizu and Komatsu 2011; Saitsu *et al.* 2019). Very recently, in order to investigate the efficacy and safety of mycelia of the fungus enriched with 5 mg/g erinacine A (**7**), a clinical trial study was conducted. Patients with mild Alzheimer's disease took 3 capsules (lyophilized mycelia 350 mg/capsule, containing 5 mg/g erinacine A) per day. This study comprised a 3-week-no-drug screening period, followed by a 49-week double-blind treatment period with 2 parallel

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Figure 2. Fairy ring that appeared on the campus of Kwansei Gakuin.

groups in which eligible patients were randomized to either 3 mycelia capsules per day or identical appearing placebo capsules. As a result, scores that measure intellectual health performance such as Cognitive Abilities Screening Instrument (CASI), Mini-Mental State Examination (MMSE), and Instrumental Activities of Daily Living (IADL) of the patients were significantly increased by intake of the capsules compared to the placebo group. This human trial was performed based on various *in vitro* and *in vivo* reports that erinacine A (**7**) had positive effects on dementia and was the first report of the antidementia effect of the mycelia of the fungus on human (Li *et al.* 2020).

Fairy-ring causing principles "fairy chemicals"

The rings, ribbons or arcs of stimulated plant growth and/or of the fruiting bodies of the larger fungi that often occur in floors of woodlands and agricultural or amenity grassland in most parts of the world are commonly called "fairy rings" (Figure 2) (Evershed 1884; Smith 1957; Couch 1995). 2-Azahypoxanthine (AHX, **20**) and imidazole-4-carboxamide (ICA, **21**) were isolated from a fairy-ring-forming fungus *Lepista sordida* (Komurasakishimeji in Japanese) (Choi *et al.* 2010a,b). AHX was converted into a metabolite 2-aza-8-oxo-hypoxanthine (AOH, **22**) in plants (Figure 3) (Choi *et al.* 2014). It was found that these 3 compounds, named as fairy chemicals (FCs), endogenously exist in plants and are biosynthesized via a new purine metabolic pathway both in plants and the fungus *L. sordida* (Choi *et al.* 2014; Mitchinson 2014; Suzuki *et al.* 2016). FCs provided tolerance to plants against various stresses and regulated the growth of all the plants tested (Choi *et al.* 2010a,b, 2014). In addition, FCs increased the yield of rice, wheat, and other crops in greenhouse and/or field experiments (Choi *et al.* 2010a,b, 2014; Tobina *et al.* 2014; Asai *et al.* 2015).

I have already published a review about "fairy rings" in this journal in 2018; therefore, here I describe the progress of the research since then (Kawagishi 2018).

Series of AHX derivatives (**23-30**) were synthesized by Pd catalyzed C–H arylation by Itami *et al.* as collaborative research with us and their growth-promoting activity against rice was evaluated (Figure 4). The activity of all the derivatives (**23-30**) tend to be stronger than **20**. The introduction of phenyl (**23**), *o*-methylphenyl (**24**), 4-biphenylyl (**27**), 1-naphthyl (**28**), or *m*-chlorophenyl (**30**) moiety at the C6 position of AHX enhanced the root growth promoting activity compared with **20**. This result indicated the possibility that chemical modification has potential for creating more active and safer FC derivatives (Kitano *et al.* 2018).

In order to investigate the biosynthetic pathway of FCs in detailed and more accurately, I thought that double-^{13}C-labeled FCs would be very useful, especially, FCs having 2 ^{13}C-labeled carbons in the imidazole ring would be ideal. Synthesis of the desired compounds was accomplished by Kan *et al.* as collaborative research with us; double-^{13}C-labeled AHX (**31**), ICA (**32**), 5-aminoimidazole-4-carboxamide (AICA, **33**), and AICA-1-β-riboside (AICAr, **34**) were synthesized from stable isotope-labeled sodium cyanide and triethyl orthoformate (Ouchi *et al.* 2018). Endogenous AHX (**20**) and AOH (**22**) had been detected by liquid chromatography-tandem mass spectrometry (LC-MS/MS); however, the detection of **21** had been very difficult due to a very low sensitivity of it by LC-MS/MS (Mitchinson 2014). We developed a high-sensitivity detection method for FCs including **21** by successive pretreatments of crude extracts with 2 columns. Using this new method, the endogenous **21** was detected in various plants and mushrooms (Takemura *et al.* 2019). The quantitative analysis of the endogenous level of **21** in rice and Arabidopsis was performed using the double-^{13}C-labeled ICA (**32**) (Takemura *et al.* 2019). In addition, the incorporation experiment of the labeled AICA (**33**) into rice and the enzyme assay using partially purified enzyme fraction of rice, **32**, and **33** indicated that all FCs (**20-22**) are biosynthesized from 5-aminoimidazole-4-carboxamide (AICA, **35**), an important metabolite on the purine metabolic pathway as a precursor of hypoxanthine, hypoxanthine, uric acid, and so on (Takemura *et al.* 2019). The gene expression profile of rice treated with **21** showed largely reverse patterns in comparison to those of **20** and **22**, suggesting that the plant growth may be regulated by the endogenous levels of the 3 FCs (**20-22**). It also revealed that the treatment with **21** imparted stress tolerance to plants like **20** and **22** (Takemura *et al.* 2019).

Two metabolites (**36** and **37**) were isolated from **21**-treated rice plant, and their structures were determined by spectroscopic analysis including the single-crystal X-ray diffraction technique and synthesis. The ribotide of ICA (**38**), whose existence was predicted by us, was also synthesized and detected from the treated rice plant by LC-MS/MS. These results indicated that rice might interconvert **21**, **36**, and **38** to regulate the physiologically activity (Choi *et al.* 2019).

We revealed that FCs are biosynthesized from the purine metabolic pathway in plants; however, the biosynthetic pathway of FCs in fungi remained less clear than that in plants. Therefore, the fungus *L. sordida* was incubated with double-^{13}C-labeled AICA (**33**) (Figure 5). As a result, **31** and **32** were detected in the culture broth. The result suggested that the fungus released **31** and **32** as metabolites of **33** out of the cells (Ito *et al.* 2020).

We were also interested in the origin of the carbon skeleton of FCs. Therefore, the fungus *L. sordida* was incubated with [1,2-$^{13}C_2$] Gly. It is well known that 2 juncture carbons in the purine carbon skeleton of all the purine bases come from Gly in the purine metabolic pathway. After incubation of the fungus with the double-labeled Gly, **20** and **21** were purified from the culture broth and both of the obtained compounds were analyzed by ^{13}C NMR and LC-MS/MS. In the ^{13}C NMR spectra, the signals of the juncture carbons, C-4 and C-5, in AHX (**20**) and ICA (**21**) were remarkably increased and coupled to each other to form doublets (Figure 6). In the LC-MS/MS analyses, unlabeled AHX (**20**) and ICA (**21**) (authentic standards) gave only native parent ion peaks (*m/z* 136 and 110, respectively) in the negative mode. In contrast,

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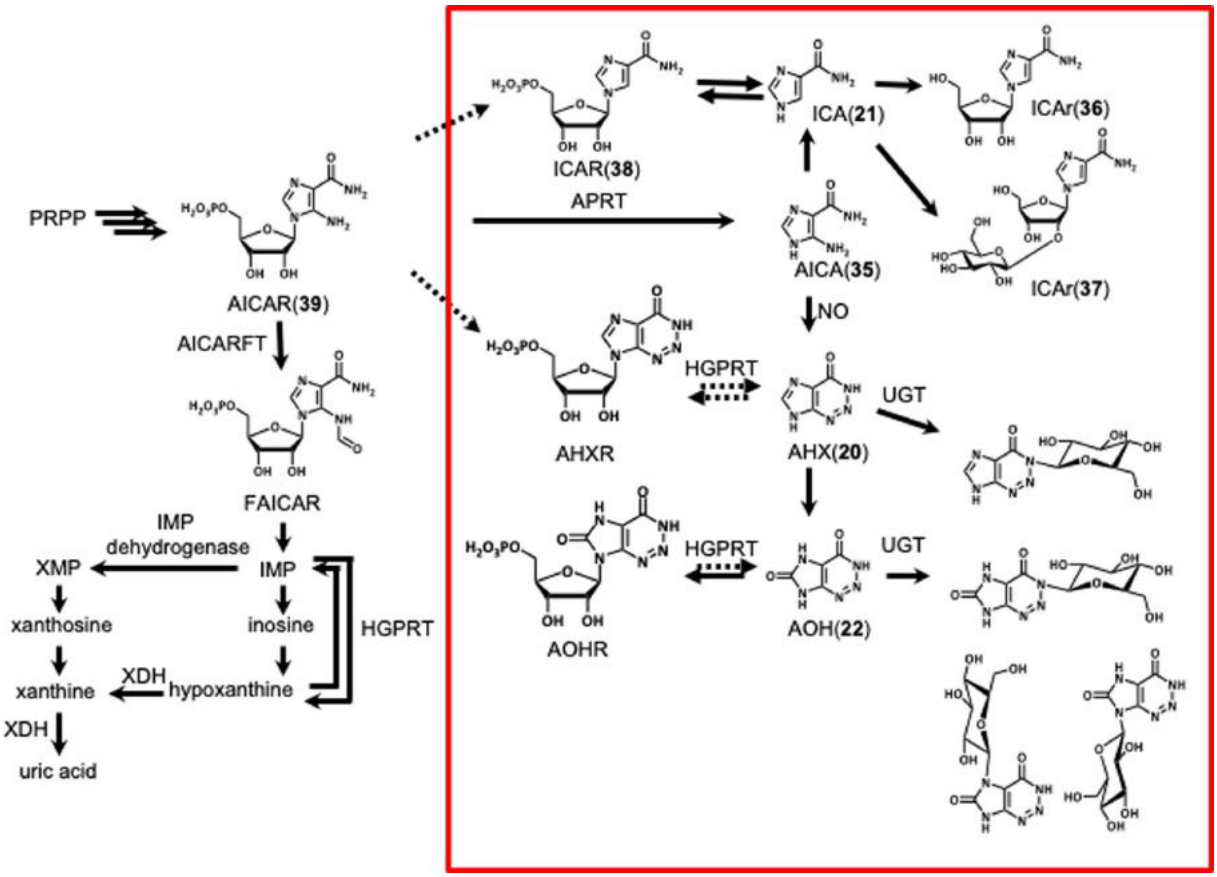

Figure 3. Purine metabolic pathway in animals, plants, and microorganisms including the novel route. The routes shown with black arrows were cited from Kyoto Encyclopedia of Genes and Genomes. New metabolites and their biosynthetic routes in plant and/or the fungus *Lepista sordida* are surrounded in the red square.

purified AHX and ICA gave parent ion peaks in which 2 carbons were derived from the labeled Gly (*m/z* 138 and 112, respectively) along with native parent ion peaks. These results indicated that 2 carbons of Gly were incorporated at the junctures of the purine-like structure in **20** and **21**, and both of the compounds were biosynthesized from the purine metabolic pathway in the fungus like the other general purine bases (Ito *et al.* 2020).

Quantitative reverse transcription-PCR analysis demonstrated that the adenine phosphoribosyltransferase gene of *L. sordida* mycelia (LsAPRT) showed transcriptional enhancement after adding of 5-aminoimidazole-4-carboxamide-1-β-D-ribofuranosyl 5′-monophosphate (AICAR, **39**). We thought that LsAPRT was involved in the conversion reaction from **39** to **35**; therefore, the gene was overexpressed in *Escherichia coli*. The recombinant enzyme *r*LsAPRT catalyzed reversible interconversion not only between **39** and **35** but also between **38** and **21**. Furthermore, the endogenous presence of **38** in the fungus mycelia was proven by LC-MS/MS analysis. The results indicated that there is a novel metabolic pathway of **21** in which **38** was an intermediate in the fungus (Figure 3) (Ito *et al.* 2020).

It is possible that chemical signal(s) that regulates fungus growth is secreted by turfgrass during development of a fairy-ring forming fungus on fairy rings. However, the molecule(s) was completely unknown. Therefore, we searched for mycelial growth regulator(s) produced by turfgrass and succeeded in the isolation of an active compound and identified it as 2-acetyl-3,5-dimethoxyphenol (**40**) from turfgrass *Agrostis stolonifera* (Figure 7). This compound inhibited mycelial growth of 2 basidiomycetes, *Flammulina velutipes* (Enokitake in Japanese) and *Coprinopsis cinerea* (Ushigusohitoyatake in Japanese), as well as the fairy ring-forming fungus *L. sordida* (Choi *et al.* 2018).

Conclusion and perspectives

Concerning antidementia compounds from the fungus *Hericium erinaceus*, there is no direct evidence that the compounds themselves discovered by us have the antidementia effects on human. At present, it would be appropriate to say that we have discovered a group of compounds that are likely to be effective against human dementia. However, it is certain that our studies triggered the worldwide research on the effects of the fungus on various brain functions, and in that regard our research may be meaningful. Nowadays, supplements derived from this fungus for the improvement of brain functions are commercially available not only in Japan but also in Europe, America, China, and Korea.

Plant hormones are signal molecules produced within plants, which occur in extremely low concentrations. Plant hormones control all aspects of growth and development, from embryogenesis, the regulation of organ size, pathogen defense, stress tolerance and through to reproductive development. Unlike in animals, each plant cell is capable of producing hormones (https://en.wikipedia.org/wiki/Plant_hormone; Kawagishi 2019). We

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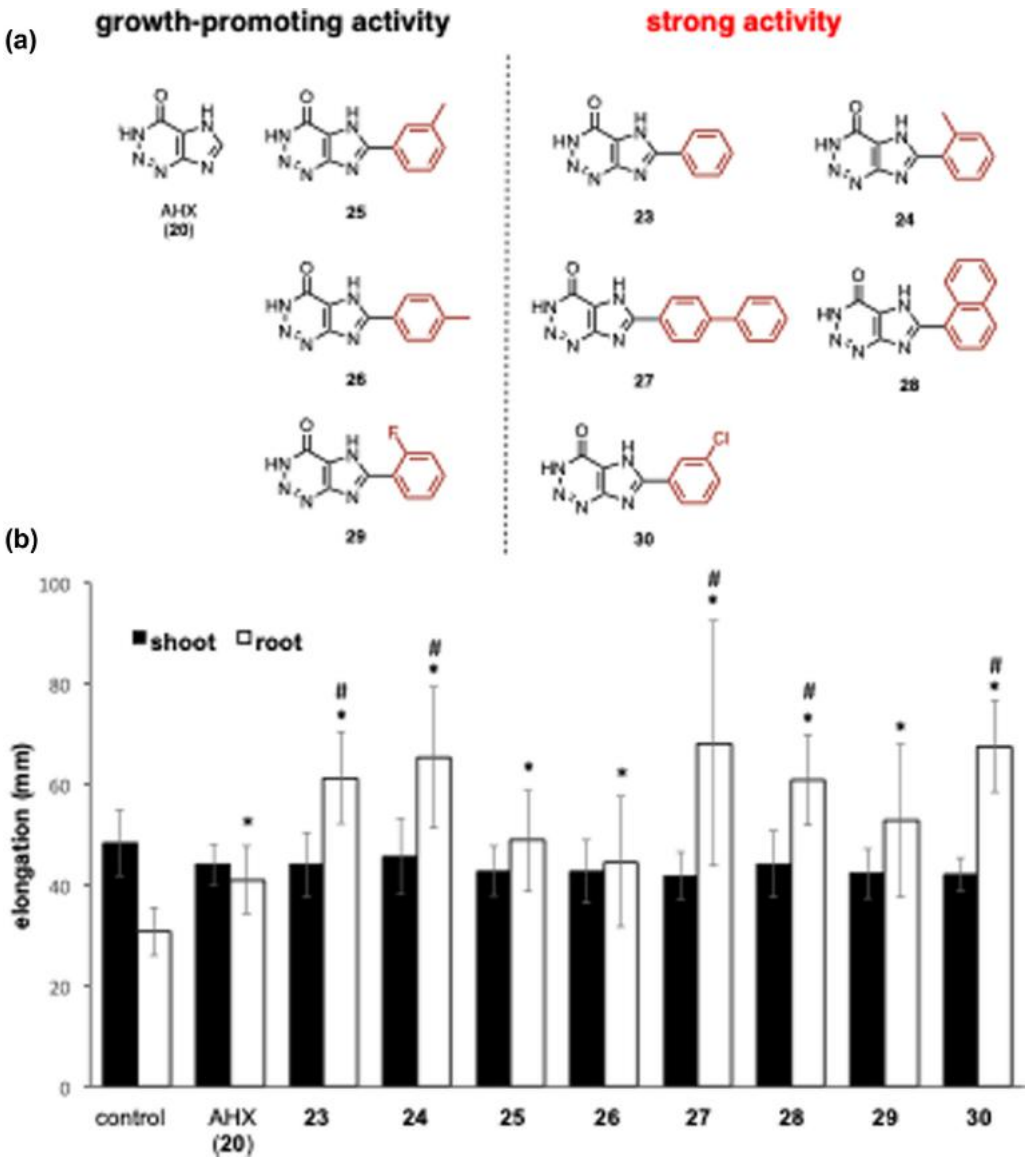

Figure 4. Effect of **23-30** on the growth of rice plant. (a) Chemical structures of **23-30**. (b) Effect on the growth of rice plant. Germinated seeds were treated with the compounds. Results are the mean ± SE ($n = 12$). Asterisk indicates a value that is significantly different form the control (Student's *t*-test, $P < .05$).

believe that FCs are a new family of plant hormones and are tracing the history of a plant hormone, gibberellins (Kawagishi 2018, 2019). Gibberellins were first isolated as toxins from fungus *Gibberella fujikuroi* that causes Bakanae disease in the rice plant and one of whose symptom is abnormal growth of rice internodes (Kurosawa 1926; Yabuta 1935). Afterward, it was proven that those compounds exist in all plants, and then they got confirmed as a plant hormone. Definition of plant hormones are "signal molecules produced within plants that occur in extremely low concentrations and control all aspects of growth and development, from embryogenesis, the regulation of organ size, pathogen defense, stress tolerance and through to reproductive development." All of our findings about FCs meet the condition for a plant hormone for now and we are now doing various trials to prove FCs as a new family of plant and fungi hormones. If FCs is recognized as a new family of plant hormones, it will be the second plant hormone original to Japan after gibberellin.

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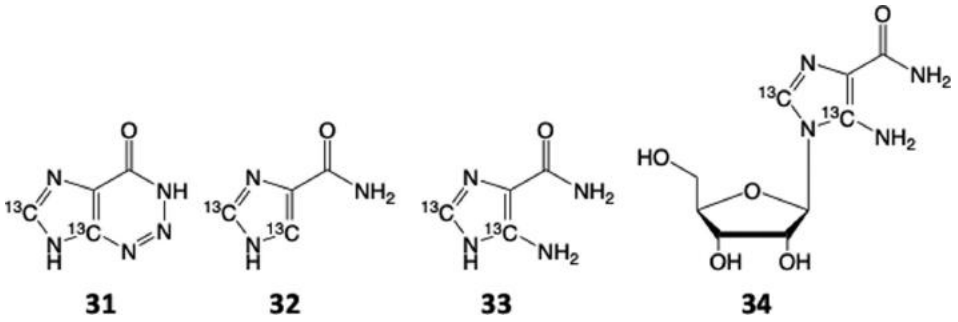

Figure 5. Structure of double-^{13}C-labeled FCs and the related compounds.

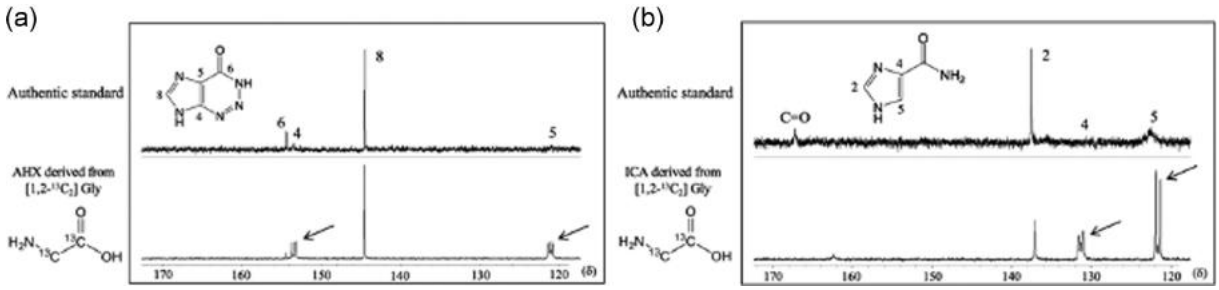

Figure 6. (A) ^{13}C NMR spectra of AHX (**20**) and (B) ICA (**21**) derived from [1,2-$^{13}C_2$] Gly in CD_3OD. Top and bottom indicate the spectra of authentic standards and purified AHX (**20**) and (B) ICA (**21**) derived from labeled Gly, respectively. Black arrows represent the signals of ^{13}C from [1,2-$^{13}C_2$] Gly.

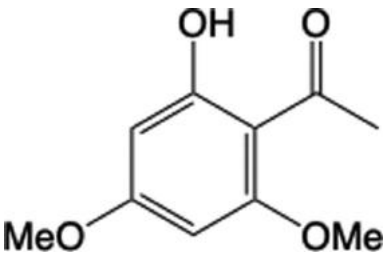

Figure 7. Structure of the mycelial growth regulator from turfgrass.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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